Tutorial Problem

Consider a rectangular wing with an aspect ratio of 6, an induced drag factor $\delta = 0.055$, $\tau = 0.055$, and a zero-lift angle of attack of −2°. At an angle of attack of 3.4°, the induced drag coefficient for this wing is 0.01. Calculate the induced drag coefficient for a similar wing (a rectangular wing with the same airfoil section) at the same angle of attack, but with an aspect ratio of 10. Assume that the induced factors for drag and the lift slope, δ and τ, respectively are equal to each other (i.e. $\delta = \tau$). Also, for AR = 10, $\delta = 0.105$, and $\tau = 0.105$.

Solution:

$$\text{Span efficiency factor} = e = \frac{1}{1+\delta} = \frac{1}{1+0.055} = 0.948$$

$$C_{D,i} = \frac{C_L^2}{\pi e AR} \Rightarrow C_L = \sqrt{C_{D,i}\pi e AR} = \sqrt{(0.01\times\pi\times0.948\times6)} = 0.4227$$

The lift slope of the wing:

$$\frac{dC_L}{d\alpha} = a = \frac{0.4227-0}{3.4-(-2)} = 0.078\,/\,\text{degree} = 4.485\,/\,\text{rad}$$

From the lift slope of the wing,

$$a = \frac{a_0}{1+(a_0/\pi AR)(1+\tau)}$$, a_0 = lift curve slope of an airfoil

τ is a function of the Fourier coefficients A_n.

$$4.485 = \frac{a_0}{1+\left[(1.055)a_0/\pi(6)\right]} = \frac{a_0}{1+0.056a_0} \Rightarrow a_0 = 5.989\,/\,\text{rad}$$

The lift slope of the modified wing is given by,

a_0 remains the same because the airfoil is fixed. Only the size of the wing has changed.

$$a = \frac{a_0}{1+(a_0/\pi AR)(1+\tau)} = \frac{5.989}{1+\left(5.989/(\pi)(10)\right)(1+0.105)} = 4.95\,/\,\text{rad} = 0.086\,/\,\text{deg}$$

The lift coefficient for the second wing is therefore

$$C_L = a(\alpha-\alpha_{L=0}) = 0.086\left(3.4-(-2)\right) = 0.464$$

In turn, the induced drag coefficient is

$$C_{D,i} = \frac{C_L^2}{\pi e AR} = \frac{(0.464)^2}{\pi\times0.905\times10} = 0.0076$$